

Resuscitation with Lactated Ringer's Solution in Rats with Hemorrhagic Shock Induces Immediate Apoptosis

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Background: We hypothesize that different resuscitative fluids may immediately affect the degree of apoptosis after hemorrhagic shock.

Methods: Rats ($n = 35$) were hemorrhaged 27 mL/kg over 5 minutes followed by 1 hour of shock, then resuscitation over 1 hour. The six treatment groups were sham hemorrhage, sham resuscitation, whole blood resuscitation, lactated Ringer's solution (LR) resuscitation with three times the volume bled, sham hemorrhage with LR infusion, and 7.5% hypertonic saline resuscitation (9.7 mL/kg). Liver and small intestine were harvested immediately after resuscitation. Apoptosis was evaluated by using in situ cell death detection method.

Results: Resuscitation with LR resulted in a significant increase in small intestinal and liver apoptosis. Animals that received LR infusion without hemorrhage had an increased level

of apoptosis in the intestine. Apoptosis in the intestine was observed in both the mucosa and muscularis externa. There was no increase in apoptosis in either organ in the animals resuscitated with sham resuscitation, whole blood, and hypertonic saline compared with the sham hemorrhage group.

Conclusion: Resuscitation with LR solution after hemorrhagic shock increased immediate cell death by apoptosis in both the small intestine and liver. There was no significant increase in apoptosis in the animals resuscitated with hypertonic saline, whole blood, or in unresuscitated animals. Thus, the type of resuscitation fluid used may affect the apoptotic cellular response to shock.

Key Words: Programmed cell death, Liver, Small intestine, Lactated Ringer's solution, Tunnel.

Successful treatment of hemorrhagic shock includes surgical control of hemorrhage and restoration of tissue perfusion. Although the delivery of intravenous fluids and blood to the patient is well established, the adverse sequela of reperfusion injury is only now being appreciated. Decreased perfusion during the shock state can injure cells, but the reestablishment of blood flow through ischemic tissue induces generalized inflammation and perpetuates the injury cycle.¹ Cellular damage from reperfusion leading to cell death and subsequent organ failure is generally accepted in trauma as the most adverse consequence of resuscitation from severe hemorrhagic shock. However, a thorough understanding of the nature of the cell injury has not yet been fully elucidated.

Cells die by two principal pathways, necrosis and apoptosis.^{2,3} Cell necrosis occurs in acute nonphysiologic injury. It is a random and passive phenomenon associated

with inflammation and is a firmly established consequence of hemorrhagic shock. Unlike necrosis, apoptosis is physiologic and highly regulated, involving gene modulation and complex pathways of signal transduction.³ The production of specialized gene products with the resulting nuclear fragmentation and eventual receptor mediated phagocytosis in the absence of inflammation are hallmarks of programmed cell death.^{4,5} Recent evidence suggests apoptotic cell death is important in the pathophysiology of trauma-hemorrhage by increasing thymic apoptosis and impairing intestinal mucosal immunity of Peyer's patches.^{6,7} Apoptosis has also been demonstrated in various situations of ischemia and reperfusion in several organs such as the brain, heart, kidney, pancreas, liver, and intestines.⁸⁻¹⁶ It seems clear that the process of programmed cell death is firmly established as an important element of reperfusion injury. Isolated ischemia-reperfusion and hemorrhagic shock resuscitation, however, are not the same phenomenon on the basis of etiology and pathophysiology. A pivotal difference is that in hemorrhagic shock the composition of the cellular milieu is a function of the kind of fluid used in resuscitation. Hemorrhagic shock also does not produce complete ischemia of tissue beds. It has previously been shown that the type of fluid used to treat hemorrhage can influence the cellular response such as the activation of neutrophils in swine resuscitated with lactated Ringer's solution (LR).¹⁷ Consequently, the establishment of programmed cell death as a repercussion of blood volume restoration with various resuscitation fluids after hemorrhage is not well defined. Therefore, we sought to determine the degree of apoptosis after resuscitation with LR after hemorrhagic shock.

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TABLE 1. Mean arterial pressures, hemoglobin levels, and end resuscitation pH^a

Groups	Mean Arterial Pressure (mm Hg)			Hemoglobin (g/dl)		pH
	Baseline	At Shock	Resuscitation	Baseline	Resuscitation	
SH	96.3 ± 10.2	81.5 ± 7.5	96.3 ± 5.5	NA	12.1 ± 0.4	7.43 ± 0.03
H-SR	90.5 ± 6.8	25.9 ± 5.4 ^b	65.8 ± 8.0 ^b	11.2 ± 0.3	9.6 ± 0.2 ^b	7.46 ± 0.04
H-WB	83.3 ± 10.8	31.1 ± 5.9 ^b	95.1 ± 7.8	11.9 ± 0.3	10.6 ± 0.4 ^b	7.47 ± 0.01
H-LR	106.3 ± 6.7	34.6 ± 10.5 ^b	74.8 ± 13.0	11.3 ± 0.4	6.7 ± 0.0 ^b	7.44 ± 0.02
SH-LR	73.3 ± 8.3	81.1 ± 5.3	83.2 ± 10.5	11.1 ± 0.0	9.6 ± 0.9 ^b	7.43 ± 0.03
H-HTS	118.1 ± 3.4	29.9 ± 10.7 ^b	89.6 ± 8.3	12.0 ± 0.3	8.0 ± 0.3 ^b	7.35 ± 0.02

^a SH, sham hemorrhage, cannulation only; H-SR, hemorrhage-sham resuscitation; H-WB, hemorrhage-whole blood resuscitation, 27 mL/kg; H-LR, hemorrhage-resuscitation at 81 mL/kg; SH-LR, sham hemorrhage-LR infusion at 81 mL/kg; and H-HTS, hemorrhage-7.5% hypertonic saline resuscitation at 9.7 mL/kg. The data are represented as mean ± SEM.

^b $p < 0.05$, compared with the sham group.

MATERIALS AND METHODS

This study was approved by the institutional Laboratory Animal Review Board for the care and use of animals. All research was conducted in compliance with the Animal Welfare Act and other federal statutes and regulations relating to animals and experiments involving animals, and adhered to the principles stated in the *Guide for the Care and Use of Laboratory Animals*, NIH publication 86-23, 1985 edition.

Male Sprague-Dawley rats (Charles River Laboratories, Inc., Frederick, Md), each weighing between 350 and 400 g, was used in this study. They were maintained on a 12-hour light cycle and were provided standard rat chow and water ad libitum throughout the study. Animals ($n = 35$) were placed under inhaled isoflurane 1 to 2% and were allowed to spontaneously breathe by using a veterinary anesthesia delivery system (ADS 1000, Engler Engineering Corporation, Hialeah, Fla). Under aseptic conditions, the right common carotid artery and external jugular vein were cannulated with sterile polyethylene catheters (PE 50, Clay Adams, Piscataway, NJ). Intravenous heparin at 10 units per mL was instilled into the catheters to prevent thrombosis. After cannulation, the animals were connected to a pressure transducer and computerized physiograph system (Buxco Electronics, Sharon, Conn) for continuous recording of blood pressure and heart rate. Arterial blood samples were analyzed at baseline, at the end of the shock period and after resuscitation by a Stat Profile 2 Blood Gas and Electrolyte Analyzer (Nova Biomedical, Waltham, Mass).

Animal hemorrhage (27 mL/kg) was through the carotid catheter over 5 minutes followed by a 60-minute period of hypovolemic shock. During the shock phase, the animals were maintained at 1% isoflurane. Animals were randomized into the following groups: sham hemorrhage ($n = 4$, SH), hemorrhage-sham resuscitation ($n = 6$, H-SR), hemorrhage-whole blood resuscitation ($n = 7$, H-WB), hemorrhage-LR resuscitation with three times the blood loss volume (81 mL/kg, $n = 6$, H-LR), sham hemorrhage-LR infusion (81 mL/kg, $n = 6$, SH-LR), and hemorrhage-7.5% hypertonic saline resuscitation ($n = 6$, H-HTS). The administered volume of 9.7 mL/kg for HTS gives the same milliequivalents of sodium as 81 mL/kg resuscitation with 0.9% normal saline. All animals were resuscitated over 1 hour by means of the jugular vein with a Harvard Apparatus Infusion Pump (South Natick, Mass). After resuscitation, the liver and distal small intestine was immediately harvested into chilled in 10% buffered formaldehyde for fixation.

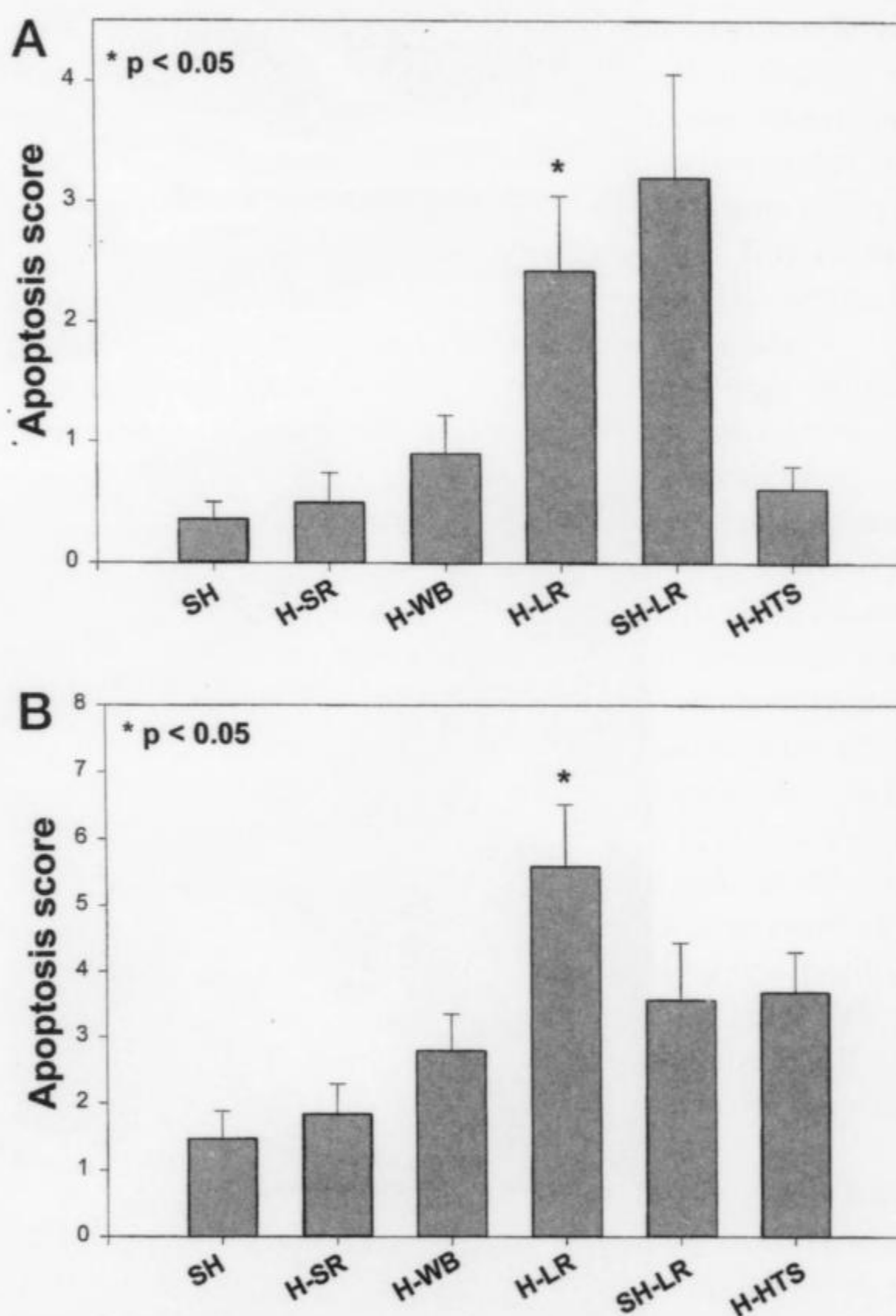


FIG 1. (A) Apoptosis in the small intestine. SH, sham (cannulation only); H-SR, hemorrhage-sham resuscitation; H-WB, hemorrhage-whole blood resuscitation, 27 mL/kg; H-LR, hemorrhage-resuscitation at 81 mL/kg; SH-LR, sham hemorrhage-LR infusion at 81 mL/kg; and H-HTS, hemorrhage-7.5% hypertonic saline resuscitation at 9.7 mL/kg. The height of each column represents the mean ± SD of the apoptosis score given to each section per group. * p less than 0.05, compared with the sham group. (B) Apoptosis in the muscularis externa of the small intestine. SH, sham (cannulation only); H-SR, hemorrhage-sham resuscitation; H-WB, hemorrhage-whole blood resuscitation, 27 mL/kg; H-LR, hemorrhage-resuscitation at 81 mL/kg; SH-LR, sham hemorrhage-LR infusion at 81 mL/kg; and H-HTS, hemorrhage-7.5% hypertonic saline resuscitation at 9.7 mL/kg. The height of each column represents the mean ± SD of the apoptosis score given to each section per group. * p less than 0.05, compared with the sham group.

Tissues were embedded in paraffin, and 5- μ m thick sections were prepared. In situ cell death detection was used to identify apoptosis in tissue sections. This method takes advantage of the DNA strand breaks that occur early in the apoptosis process.²

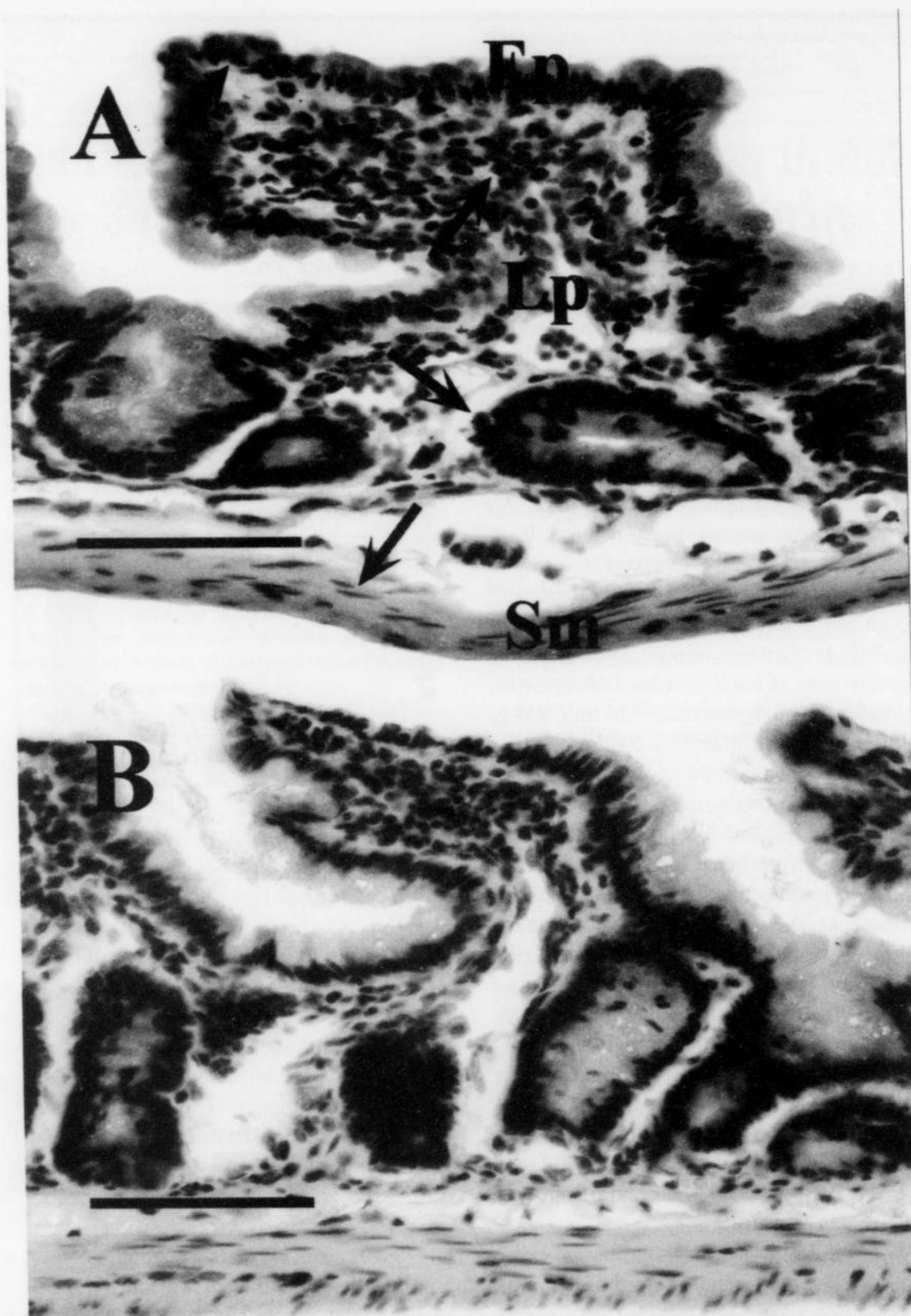


FIG 2. LR enhanced apoptosis in the gut. (A) Positive cells (arrows) are seen in the mucosa and muscularis externa of a H-LR animal. (B) SH animal with a few positive cells at the villus tip. Sm, smooth muscle cell; Ep, epithelium; Lp, lamina propria. Scale bars = 100 μ m.

Tissue sections were deparaffinized with histologic grade xylene and rehydrated in standard phosphate buffered saline. Sections were then treated with trypsin digestion in humidified chambers at 37°C for 50 minutes. They were then permeabilized with 1% Triton X-100 in buffered saline before incubation with terminal deoxynucleotidyl transferase (Boehringer-Mannheim Corp., Indianapolis, Ind) for 60 minutes at 37°C. This enzyme catalyzes the polymerization of nucleotide polydeoxyuridine triphosphate (d-UTP) to the free 3'-OH groups of DNA strand breaks and preferentially labels cells undergoing apoptosis in contrast to necrosis. Fluorescein incorporated to the d-UTP (Boehringer-

Mannheim Corp.) moiety was then detected by anti-fluorescein antibody Fab fragments from sheep conjugated with alkaline phosphatase (Boehringer-Mannheim Corp.) after incubation at 37°C for an additional 30 minutes. Addition of the enzyme substrate Fast Red TR/Naphthol AS-MX phosphate (Sigma Chemical Co., St. Louis, Mo) to the sections resulted in dark red staining of nuclei of individual apoptotic cells. Slides were counterstained with hematoxylin blue for 2 minutes then washed and mounted for light microscopy examination. Both positive and negative control reactions were performed for both small intestine and liver sections. All sections were processed in the

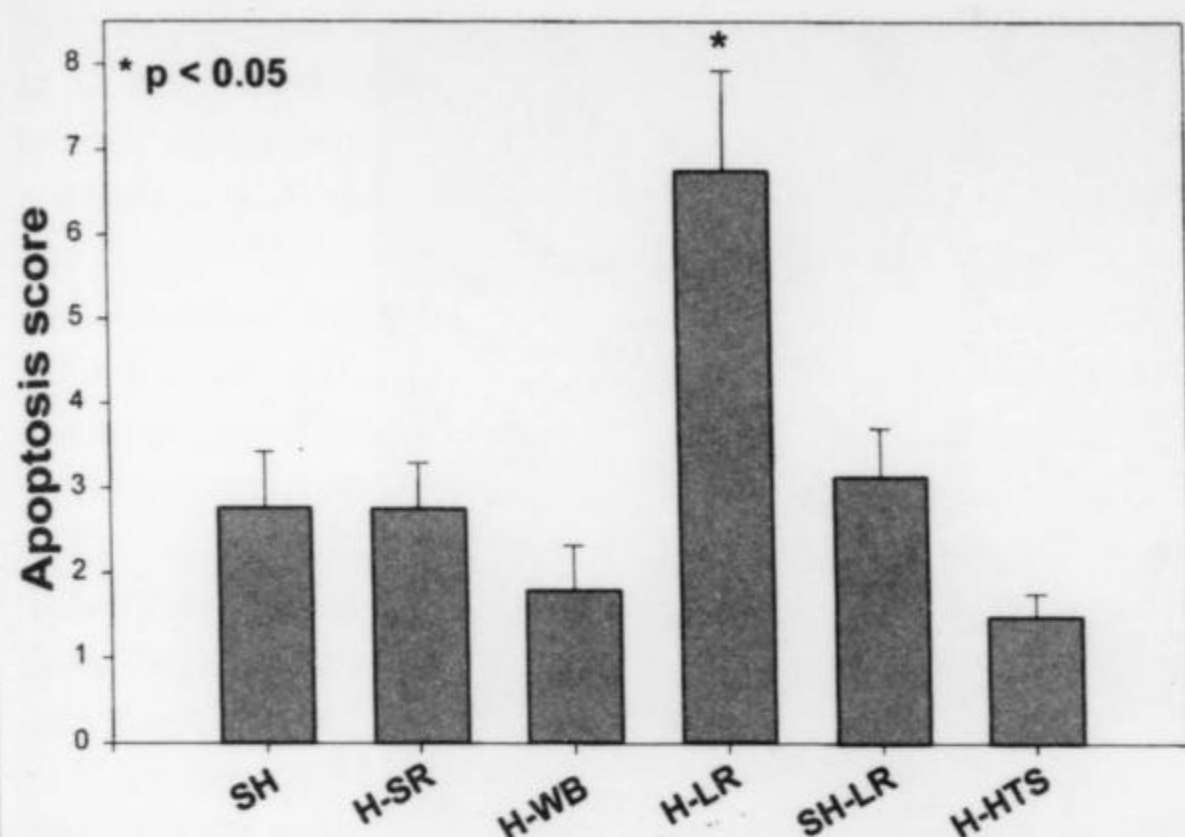


FIG 3. Apoptosis in the liver. SH, sham (cannulation only); H-SR, hemorrhage-sham resuscitation; H-WB, hemorrhage-whole blood resuscitation, 27 mL/kg; H-LR, hemorrhage-resuscitation at 81 mL/kg; S-LR, sham hemorrhage-LR infusion at 81 mL/kg; and H-HTS, hemorrhage-7.5% hypertonic saline resuscitation at 9.7 mL/kg. The height of each column represents the mean $\pm$ SD of the apoptosis score given to each section per group. * p less than 0.05, compared with the sham group.

same experiment to avoid technical artifacts and allow accurate comparison. Each tissue specimen contained three to four sections per slide for better representation of the entire organ from each animal. In addition, each tissue specimen was stained in duplicate.

Tissue sections were evaluated for the extent of positive staining and the location of stained cells relative to tissue anatomy. Two evaluators, blind to the protocol, independently examined all sections by light microscopy. Positive staining on each individual tissue section per slide was quantified with a score between 1 and 16, which was the product of the extent of apoptosis and the percentage of tissue involved. For grading the extent of apoptosis, the following scale was used: 1, $\leq 25\%$ cells staining positive; 2, $> 25\%$ but $\leq 50\%$ cells staining positive; 3, $> 50\%$ but $\leq 75\%$ cells staining positive; and 4, $> 75\%$ of cells staining positive. The percentage of the tissue section that contained positive cells was scored as follows: 1, $\leq 25\%$ of section; 2, > 25 but $\leq 50\%$ of section; 3, > 50 but $\leq 75\%$ of section; and 4, $> 75\%$ of section. The final score given to each section was the product of the apoptosis score multiplied by the number representing the area of the section involved. Evaluation of the small intestine was further divided into staining in the mucosa (epithelium, lamina propria, and muscularis interna) and staining in the bowel wall (muscularis externa). Thus, each individual tissue section from the small intestine received two scores by each reader. In the liver because of its relative homogeneity, a single number between 1 and 16 as defined in the previous text was used for each tissue section. Additionally, the zonal distribution of the majority of positive cells was recorded (centrilobular, sinusoidal, periportal, and subcapsular) for each section in the liver. All data are presented as mean $\pm$ SEM. Statistical analysis was performed with the Mann-Whitney U test for the examination of apoptosis between groups in the liver and intestines. Analysis of variance with Tukey's b test was used for comparisons of parametric data between groups. Sham hemorrhage group

was used as a control comparison group. All p values of less than 0.05 were considered significant.

RESULTS

Our current hemorrhage level of 27 mL/kg represents approximately a 40% blood volume loss based on animal body weight. All groups hemorrhaged had a precipitous drop in the mean arterial pressure (MAP) after hemorrhage. The group that underwent sham hemorrhage with LR infusion had an increase in MAP from 73.3 ± 8.3 mm Hg to 81.1 ± 5.3 mm Hg. At the end of resuscitation, all the groups had a MAP similar to the sham group except for the H-SR group, which was significantly lower than the other groups because they were not resuscitated (Table 1). Baseline hemoglobin levels were similar between groups and decreased at the end of resuscitation in all groups with the H-LR animals experiencing the largest decrease. The serum pH at baseline and at end resuscitation were comparable in all treated animals and was not statistically different from sham animals. This finding suggests no differences between groups in the degree of resuscitation by the various fluids used in the study.

In the small intestine, the group resuscitated with LR after hemorrhage, H-LR group, had a significant increase in the extent of apoptotic cells in the mucosa and smooth muscle of the bowel wall in comparison with the sham animals ($p < 0.05$, Figs. 1, A and B). The H-SR, H-WB, and H-HTS groups were not significantly different from the sham animals in apoptotic staining in both the mucosa and the bowel wall. Animals in the SH-LR group had high levels of staining in the mucosa and the bowel wall but was not statistically significant. The distribution of positive-stained apoptotic cells in the H-LR animals included the epithelial cells of the villus and submucosal glands and the smooth muscle cells of the muscularis externa (Fig. 2A). This distribution is in contrast to that found in the SH animals, in which only a few positive-staining cells were identified in the mucosa (Fig. 2B). A similar pattern of distribution of apoptosis as the H-LR animals was identified to a much lesser extent in the other groups.

As in the intestine, the liver in the H-LR group had a significantly higher extent of positive staining in comparison with SH animals (Fig. 3). Similar levels of apoptosis as the SH group was observed in the remainder of the groups. Histologic anatomy revealed most of the positive staining to involve the parenchymal hepatocytes around the central vein as well as cells in the sinusoids of the liver lobule. Hepatocytes in the subcapsular and periportal areas were also positive in the H-LR-treated animals, although the magnitude of staining was less than the centrilobular zone. Within the periportal area, the endothelial cells of the vasculature, the cells of the bile ducts and in the sinusoids were intensely positive (Fig. 4A). This distribution of staining in the liver was not observed in the sham hemorrhage group (Fig. 4B).

DISCUSSION

Our study demonstrated that rats subjected to hemorrhagic shock sustain an immediate increase in apoptosis in both the

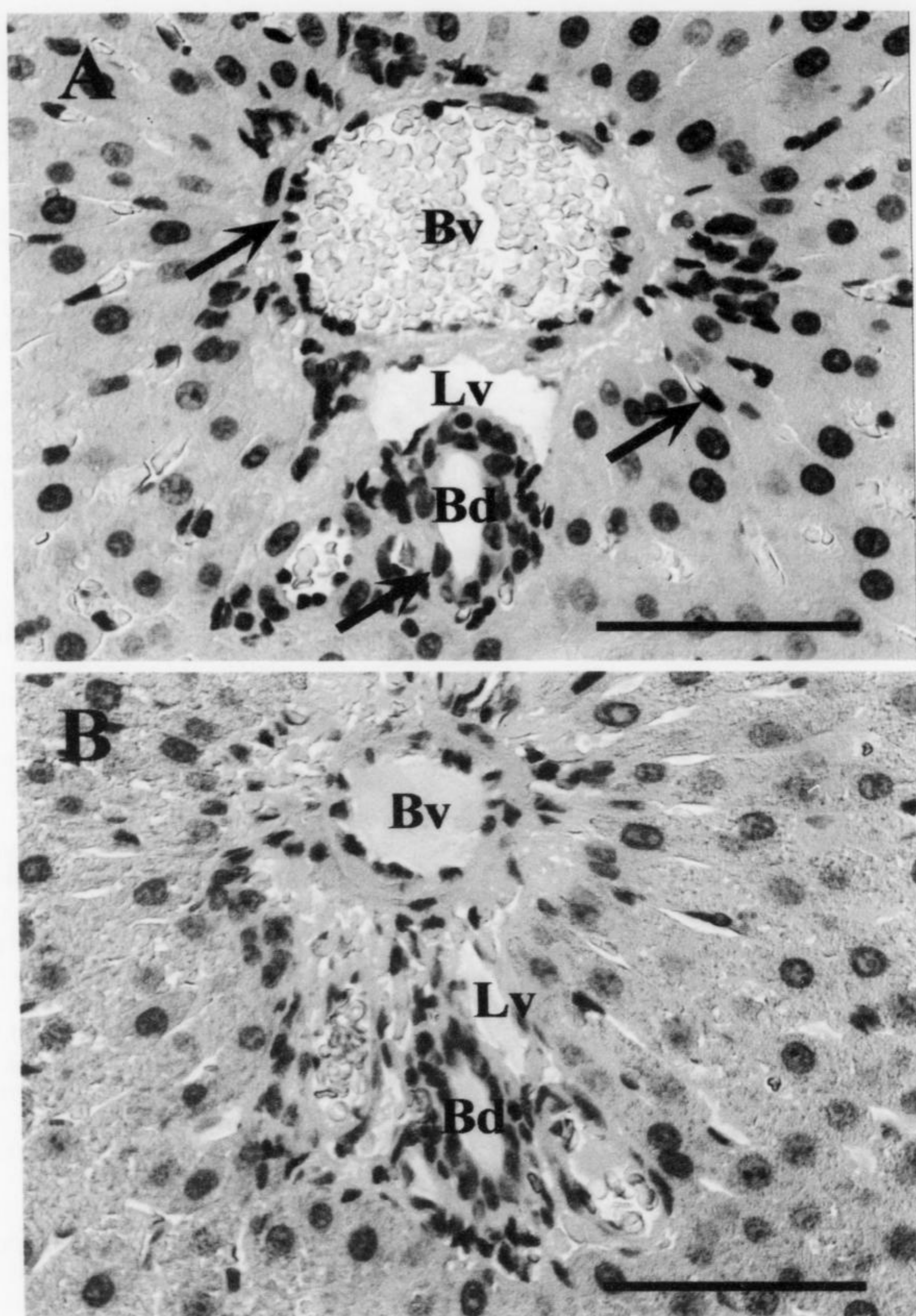


FIG 4. LR enhanced apoptosis in the liver. (A) Positive cells (arrows) are visible in the blood vessel (Bv), lymph vessel (Lv) and bile duct (Bd) in addition to parenchymal cells in an H-LR animal. (B) SH animal showing a lack of positive staining (control of A). Scale bars = 50 μ m.

liver and small intestine after resuscitation with lactated Ringer's solution. In the small intestine, the positive staining cells are located in both the mucosa and the bowel wall. Mucosal injury of the intestines by apoptosis in ischemia and ischemia-reperfusion has been described and is thought to be a major mode of cell death.^{18,19} Our findings correlate with previous studies in mucosal apoptosis; however, the extensive smooth muscle cell apoptosis of the muscularis externa we observed in the H-LR group has not been previously reported. Smooth muscle is capable of programmed cell death

as characterized in other disease processes and are known to be synthetically active for pro-apoptotic proteins.^{20,21}

Examination of the liver revealed the highest aggregate of apoptosis in the animals resuscitated with lactated Ringer's solution. The distribution of apoptotic cells was not random and exhibited a predilection for certain anatomical areas with the majority of positive staining in the hepatocytes around the central vein. Due to greater hypoxia during hypotension, the hepatocytes in this area may be more sensitive to secondary injury during reperfusion. In contrast, the ischemia-prone

subcapsular area had less staining than the centrilobular area in all the groups examined. The zonal differences observed in hepatic apoptosis after LR resuscitation in our study are consistent with the pattern of ischemia-reperfusion injury in the liver.^{13,14} This paucity of subcapsular staining in the H-LR animals may be due to less ischemia in our hemorrhage model or the LR insult may exacerbate susceptible areas in the liver. Involvement of diverse cell types, such as those in the bile ducts, blood vessels, and sinusoids indicates multiple cells residing within the liver reacted to the LR resuscitation. Biliary epithelial cells, endothelial cells, and Kupffer cells have been known to undergo apoptosis after reperfusion injury to the liver.¹²⁻¹⁴

Both whole blood and 7.5% HTS resuscitation did not increase the amount of apoptosis observed in either the liver or intestines when compared with sham animals. Although the animals in the SH-LR had similar levels of apoptosis as the SH group in the liver, in the intestines, the mucosa of the LR without hemorrhage group had a high expression of positive cells, although this level did not reach statistical significance. The SH-LR group indicates that LR by itself is adequate to increase the amount of cell apoptosis observed but that a period of low flow in combination with LR resuscitation results in the maximal amount of apoptosis. Animals in the H-SR had minimal apoptotic staining, which highlights that shock alone may not increase apoptosis and that ischemia from hemorrhagic shock is different from total ischemia-reperfusion models. This study was not designed to examine the mechanism behind the various fluids and their affect on apoptosis. In the case of LR causing increased apoptosis after resuscitation, it is probably not what is in LR but what is not in LR. Whole blood offers many advantages, including free radical scavengers and normal vascular milieu, although whole blood is not always available in trauma nor is it always needed. Research efforts should attempt to find a resuscitation fluid that is as effective as whole blood. Simple restoration of blood pressure with LR may not be adequate in situations in which very large volumes are required if we are going to attempt to reduce subsequent injury from hemorrhagic shock resuscitation. The lower levels of apoptosis in the liver and small intestine in the H-WB and H-HTS groups may indicate that reperfusion per se is not detrimental but may depend on various fluids and implicates LR as not being innocuous. The apoptosis recorded in the sham animals most likely represents the basal levels of apoptosis in the eu-volemic state.

Programmed cell death at 1 hour ensuing ischemia-reperfusion in the small intestine and within 3 hours of reperfusion in the liver has been reported by others.^{12,15,16,18} In another study, using DNA fragmentation methods, necrosis and apoptosis was found within 1 hour in mouse liver subjected to 30 minutes of warm ischemia.²² Indeed, recent reviews in the surgical literature report the process of apoptosis as relatively rapid, reaching completion within 2 hours.²³ Unlike necrosis, which can leave evidence of inflammation and tissue scarring, apoptotic cell death may not leave any visible remains, and, thus, may be missed on later examination.

Reperfusion with LR and other fluids have been shown to

up-regulate cell adhesion molecule expression rapidly after resuscitation in the lung and spleen.²⁴ This study along with our experiment show that the cell's response to resuscitation fluid involves multiple organ systems. This process is active, as necessary to synthesize de novo proteins for cell adhesion or the activation of the intracellular death program.

This research was designed to determine the immediate effect on apoptosis, in a standard volume hemorrhage shock model with standard resuscitation methods. Our data represent early cell reactions to their external milieu, occurring within the hour after the start of resuscitation. Further studies will need to be undertaken to determine the long-term functional results of the apoptosis observed in the LR resuscitated animals. Because the barrier against enteric microbes depends on intestinal mucosal integrity and the filtering capacity of the liver, it is conceivable that the extensive apoptosis in the H-LR animals may lead to increased propensity for bacterial translocation to the blood. Once considered life saving, one must now consider the effects of the resuscitation fluid on further cell injury and organ dysfunction when treating the trauma patient. Clearly, massive apoptosis of any organ will create tissue loss and functional impairment.

In conclusion, our study has demonstrated that the increase in apoptosis after hemorrhagic shock resuscitation is an early event and that this phenomenon varies with the type of resuscitation fluid used. All resuscitation fluids may not be innocuous, and the type of resuscitation fluid used may contribute to the pathologic syndromes seen after aggressive resuscitation following severe hemorrhagic shock.

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DISCUSSION

Dr. James W. Holcroft (Sacramento, California): I enjoyed the presentation very much. It was to the point and concise, as was the manuscript.

I have three main questions, and I should state at the outset that some of these questions probably are not fair. Nonetheless, I would like you to speculate a little bit. First, do you think tonicity plays a role in your findings? Lactated Ringer's is a slightly hypotonic solution; shed blood is an isotonic solution; and hypertonic saline is hypertonic. Can you draw any conclusions about the effects of the tonicity per se?

Second, do you think the results could be a function of adequacy of resuscitation? That is, could it be that some of the groups were just better resuscitated? You may have some feeling for that on the basis of blood pressure measurements or other indicators of resuscitation, such as hematocrit values to give you an idea of plasma volume expansion.

Third, do you think the results were a function of the components of the solution? I think that you are saying that lactated Ringer's is bad and it is the lactate part that is bad. Or is it possible that the chloride moiety in the chloride solutions is good?

I would like to thank Dr. Deb for sending the manuscript to me way in advance. It was extremely well written and I think the experiments are provocative and important. It was a pleasure to discuss the paper.

Dr. Martin K. Angele (Providence, Rhode Island): I enjoyed your presentation very much, and I have one comment. In your study, different volumes of various resuscitation fluids were administered. Studies from our laboratory indicate that slower rate of resuscitation following trauma-hemorrhage were associated with improved immune responses. Thus, the different rates of resuscitation in your study might contribute to the observed differences in the apoptotic frequency. Could you please comment on this?

Dr. Walter L. Biffl (Denver, Colorado): I wondered in your apoptosis scoring system if this was somehow a fraction of cells with apoptotic morphology, or if it could be that bowel wall edema played some part in altering the appearance of the bowel and subsequently the scoring of the specimen?

Dr. Basil A. Pruitt, Jr. (San Antonio, Texas): I wonder how you account for such striking variation in apoptosis in the smooth muscle of the gut and in the liver? There is a twofold difference in apoptosis in those two organs in the animals that received hypertonic salt.

There were some organs in the animals that received lactated Ringer's, after sham hemorrhage, that showed a marked increase in apoptosis score, and in other organs, there was little, if any, increase in that score. That suggests to me that there is some uncontrolled variable in action here. Perhaps it is LPS—how did you control your test system for LPS, variations of which might well cause alterations in apoptosis?

Dr. Subrato Deb (closing): Thank you for those outstanding questions.

Regarding the question of the mechanism and its relationship to tonicity, we have looked into this in our lab by varying the tonicity of the lactated Ringer's. We have examined this in both the rat and swine hemorrhagic shock models, and found no correlation in terms of neutrophil activation and the tonicity.

We have also altered the levels of the other electrolytes in LR, such as the potassium and calcium and found results that correlate with our previous studies.

In terms of the adequacy of resuscitation, we tried to demonstrate that all the treatment groups were resuscitated equally by measuring global parameters, such as mean blood pressures following resuscitation. We also measured serum pH as an indicator of volume status and found no significant differences between groups. From this data, we feel that the differences we see in apoptosis between groups are not due simply to the adequacy of resuscitation. In a separate experiment, we have looked at LR and neutrophil activation in the swine model where we placed pulmonary artery catheters and measured cardiac indices, oxygen delivery and consumption, and oxygen extraction ratios to ensure adequate resuscitation among animals and found similar results to what I presented today.

I do not know what components in LR may be activating. We speculate from work in our lab that maybe LR is lacking

in some component, such as oxygen scavengers, that may be present in fluids such as whole blood that could account for the effects we are observing.

Since we do not know, further studies, focusing more on colloids and whole blood, would be interesting to look at, and this is an ongoing project in our lab currently.

In terms of the different volumes over 1 hour that was delivered to the animals, we have looked at this phenomenon, again in the swine model, and do agree that the rate of volume delivery can affect the results. We allude to this briefly in the paper. As we gave the entire LR volume over 1 hour, we added the LR without hemorrhage to determine whether the large volume itself, instead of simply LR, could affect the results. And as seen from our data, there were no statistically significant increases in apoptosis in either the liver or the small intestine in animals given only LR and no hemorrhage.

In the apoptosis scoring question, we used a system to quantitate apoptosis by two blinded evaluators. Each slide was evaluated in terms of apoptotic cell density by looking at individual cells and the entire tissue and given a score between 1 and 16. We also looked at the histologic distribution of the apoptotic cells in each tissue section. To answer your

question, bowel wall edema did not affect our ability to analyze for apoptosis histologically.

It is difficult to measure apoptosis because it does not occur in a patchy fashion such as necrosis, rather individual cells will either be positive or negative. We found that the method of measurement we used was the best way to quantitate the amount of apoptosis in each section. Recently, we have measured bax protein expression as an indicator for apoptosis, and found identical results to what I presented today.

The variation in apoptosis between the liver and small intestine probably represents differential susceptibility of tissues to apoptotic stimuli. We do not know why the LR without hemorrhage had a high degree of apoptosis in the mucosa. This was not statistically significant due to variation. As I alluded to earlier, this may be due to the volume and rate of resuscitation. We do not know at this time.

In terms of endotoxin contamination, I can tell you that we used sterile cannulae, sterile tubing, and aseptic surgical technique. All our fluids were hospital grade, but I cannot guarantee there was no LPS contamination. I do not think endotoxin is why we are seeing differences in the degree of apoptosis between groups in our study.